INTRODUCTION

The general guidelines within this document are intended solely for the purpose of supporting Field Replacement Unit installations (FRU's). Performing any service or maintenance will always begin with the requirement to safely de-energize the main disconnect panel and utilize the Lock-Out/Tag-Out features included in the design. FRU's are defined as any component or combination of components that make up in whole or in part, a circuit within the main disconnect panel. The guidelines herein, are not to be applied as instructions and are for reference purposes only. The manufacturers installation instruction(s) and documentation provided with each individual component of the FRU must be applied in each instance. When working in, on, or near the main disconnect panel it should be treated as though it were energized at all times.

STATEMENT

ONLY a qualified technician who is knowledgeable and experienced with high voltage shall perform any service, testing, or maintenance related to the replacement of any components. All safety precautions, rules, regulations, requirements, and governing guidelines must be followed at all times when replacing and/or testing any FRU.

DISCLAIMER

This Document, the contents, and all related material herein is strictly for reference purposes only by qualified and authorized personnel. When replacing or servicing any component within the main disconnect panel refer to the specification and instructional documents for each component.

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SAFETY, DANGER, WARNING, AND CAUTION

Danger:

Only a qualified technician knowledgeable and experienced with electrical applications, systems, and installations shall perform any service or maintenance on the main disconnect panel or its parts

Danger:

The main disconnect panel should be de-energized and all Lock Out/Tag Out requirements shall be applied at the time of service to the main disconnect panel

Warning:

Always observe all safety precautions, rules, regulations, requirements, and governing guidelines when working in, on, or near the main disconnect panel

Warning:

The circuits within or parts of the circuits within the main disconnect panel may automatically re-energize once facility power or power to the main disconnect panel is restored

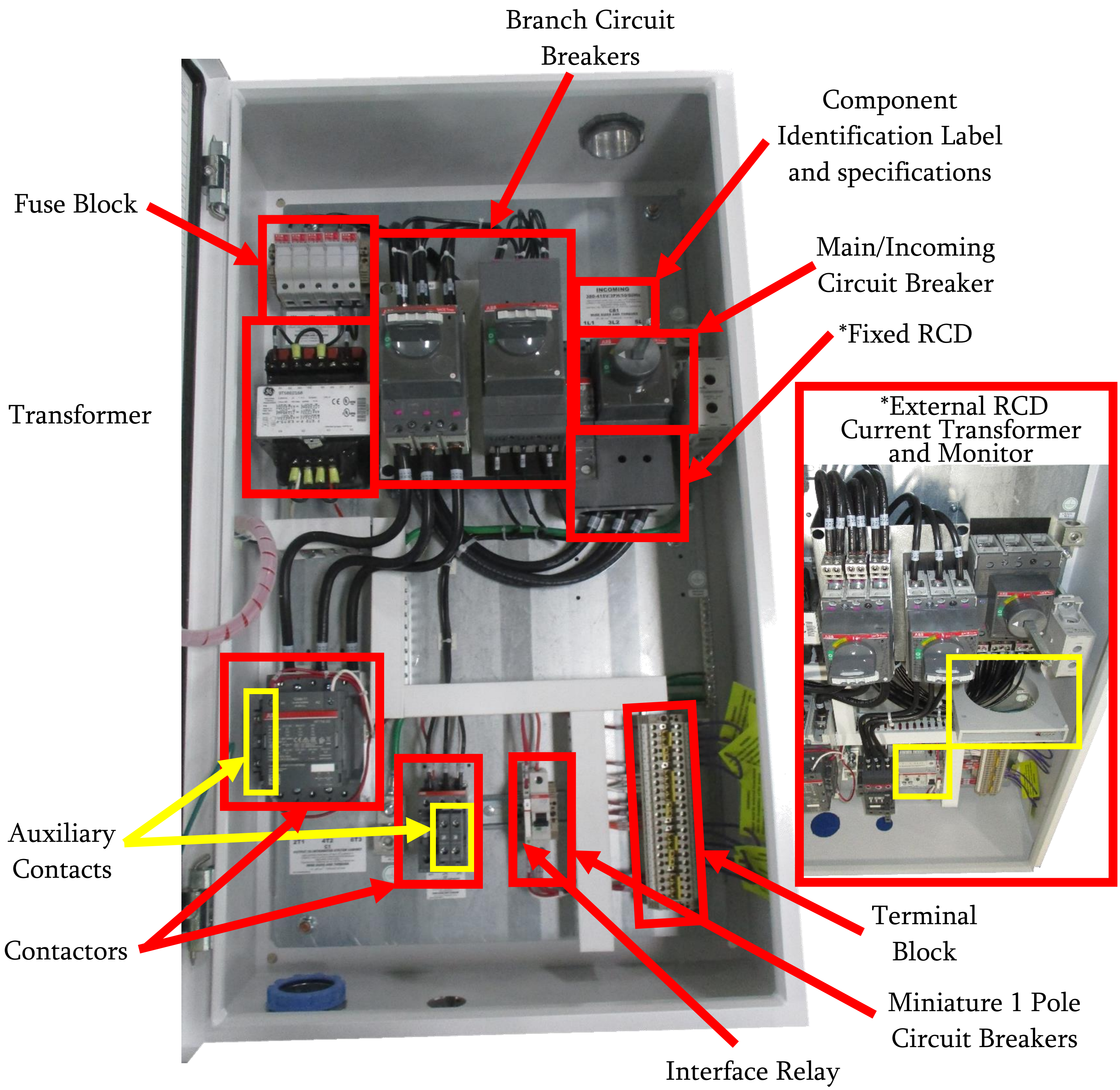
DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

DO NOT discard any hardware when replacing any component(s)

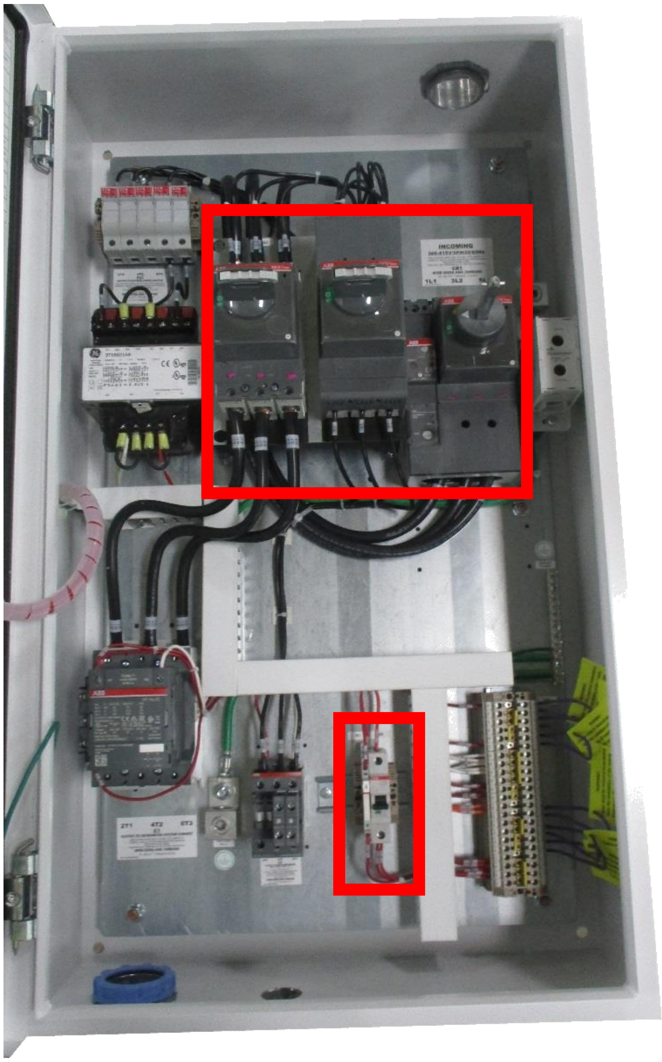
IDENTIFICAITON OF TYPICAL PANEL

The design, components, and sub components for each main disconnect panel will vary based on many factors including but not limited to manufacturer, its intended market and applicable standards, the intended design of the, and installation requirements. Below is a typical example of component locations.



CIRCUIT BREAKER FRU's

Circuit breakers within a main disconnect panel can be defined as a Main/Incoming circuit breaker, Branch/Secondary circuit breaker, or Control/Miniature circuit breaker. Always adhere to the requirements specified for torque values, acceptable wire size, and wire length. Main and Branch circuit breakers most commonly use molded case and can be equipped with numerous accessories including Operating Mechanism, RCD's, Shunt Trip, and/or Under Voltage Trip. The following examples outline the procedures to replace a main or branch circuit breaker and accessories.



The Circuit Breakers provided in this installation guide are for reference only. Each Circuit Breaker may vary from model and manufacturer. Refer to the specific instructions and specifications document for each Circuit Breaker.

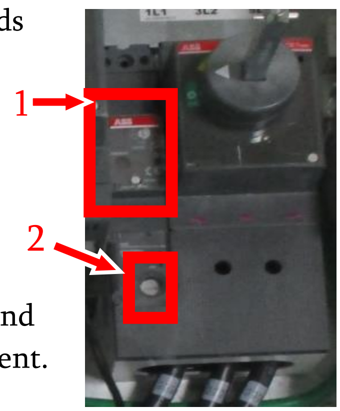
DO NOT discard any hardware when replacing any component.

MAIN AND SECONDARY CIRCUIT BREAKER FRU's

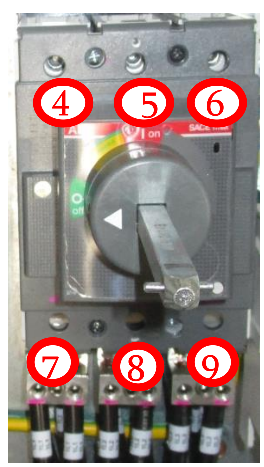
DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) and steps required to loosen the circuit breaker lugs. ***DO NOT DISCARD any hardware when replacing components***. The lugs may be covered or protected by accessories or finger guards which will need to be removed before accessing the terminals.

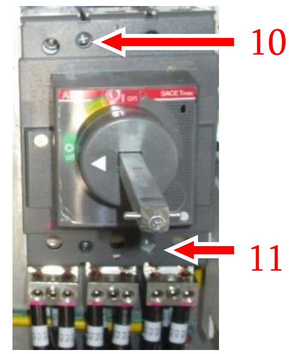


2. Prior to performing any work observe the circuit breaker configuration; RCD settings (1), wire labels (2), and termination of each component. This information can also be verified by consulting the schematic located on the inside of the enclosure door.
3. Loosen the line and load side (4-9) Lugs and remove the wires from the component. Preserve the wire bend radius and routing as much as possible.



4. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the mounting bolts

5. Loosen and remove the Mounting bolts (10-11), ***DO NOT DISCARD***, then remove the circuit breaker from the subpanel.



6. Verify the circuit breaker part number and lugs provided in the FRU kit are an exact match of the components removed from the panel.
7. Mount the circuit breaker provided in the FRU kit and install the mounting bolts from step 5.
8. Verify that the orientation of the operating mechanism is set to the OFF position and that the (2) latching pins are horizontal as shown in figure A

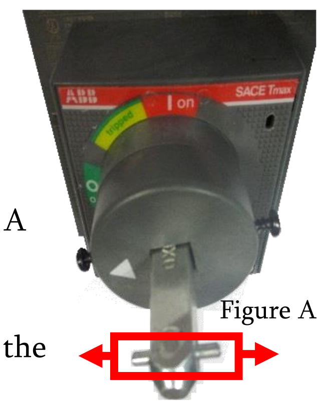


Figure A

9. As much as possible preserve the wire bending radius as noted in step 3 and then land the appropriate wires in the designated terminals on the circuit breaker.
10. Refer to the schematic on the main disconnect panel and torque each lug according to specification.

Additional instructions for Operating Mechanisms installation provided on the following page

OPERATING MECHANISM/DISCONNECT HANDLE FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) and steps required to loosen the circuit breaker face plate and operating mechanism. ***<u>DO NOT DISCARD any hardware when replacing components</u>.***

2. Remove the (2) face plate mounting screws from the circuit breaker frame. Then remove the face plate cover as shown in Figure A

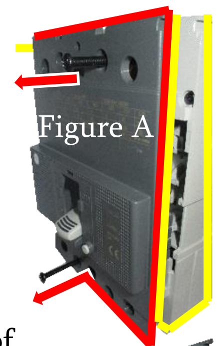


Figure A

3. The Operating Mechanism is fixed the circuit breaker face plate from the inside of the cover, by (4) mounting screws as shown in figure B. Remove all (4) screws and the operating mechanism.
4. Verify that the part number of the operating mechanism provided in the FRU kit is an exact match of the component removed from the circuit breaker.

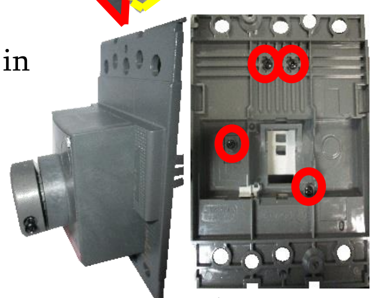


Figure B

5. Mount the operating mechanism provided in the FRU kit as shown above then fasten the operating mechanism to the circuit breaker face plate using the (4) hole locations shown in step 3.
6. Proper seating of the operating mechanism and the position of the circuit breaker lever are critical to proper functionality of the component.
7. Set the circuit breaker lever and the operating mechanism to the OFF position then align the circuit breaker lever to engage the slotted actuator of operating mechanism as shown in figure C.

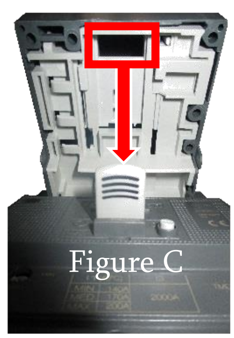


Figure C

<u>DO NOT OVER TIGHTEN</u>

8. Once the Operating Mechanism has been properly seated to the circuit breaker cover, fasten the (2) screws through the circuit breaker mounting plate as directed in step 2.
9. With the circuit breaker mounting plate and the operating mechanism secured manually test the circuit breaker for functionality by opening and closing the circuit breaker and verifying the components are secure.

<u>Additional instructions for RCD installation provided on the following page</u>

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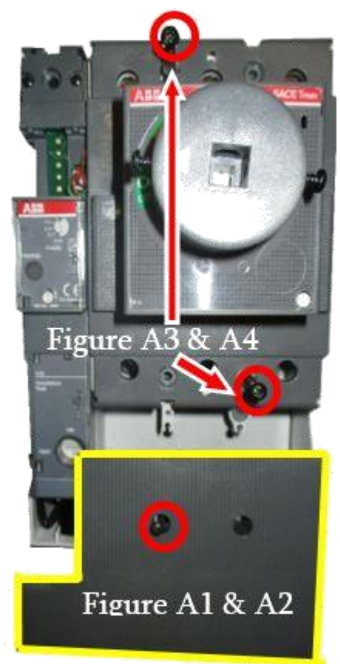
DIRECT MOUNT RCD FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) and steps required to loosen the circuit breaker frame and lugs.
2. Prior to performing any work observe the circuit breaker configuration; RCD settings (1), wire labels (2), and termination of each component.
3. The direct mount RCD is fixed to the frame of the circuit breaker and allows for the conductor to be routed from the bottom of the circuit breaker through the CT.
4. (1) Mounting hole is provided to secure the RCD phase cover to the RCD and (2) mounting holes are provided to secure the frame of the circuit breaker to the subpanel.
5. Remove the single screw and phase cover as shown in figure A1 and A2.
6. Loosen the line and load side lugs and remove the wires from the terminals as much as possible preserve the wire bending radius.
7. Remove the (2) mounting screws from the circuit breaker frame as shown in figure A3 and A4 then remove the circuit breaker and RCD from the panel.
8. Verify that the part number of the circuit breaker and RCD provided in the FRU kit is an exact match of the components removed from the panel.
9. Mount the circuit breaker and RCD using the same hardware from steps 7.
10. As much as possible preserve the wire bending radius as noted in step 6 and then land the conductors on the line and load side.
11. Reference the schematic fixed to the panel door for torque settings for each connection.
12. Replace the phase cover using the hardware in step 5



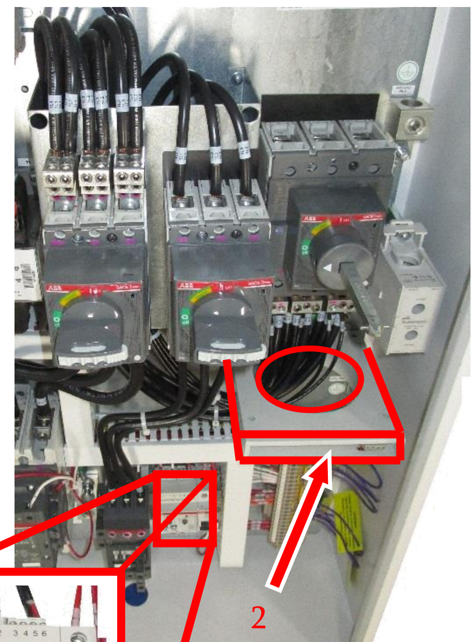
Additional instructions for RCD installation provided on the following page

EXTERNAL MOUNT RCD FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) and steps required to loosen the circuit breaker frame and lugs.
2. Prior to performing any work observe the circuit breaker configuration, wire routings, RCD settings, wire labels, and terminations for each component.
3. The external mount RCD consists of a CT (2) and RCD module (1). Similar to the Direct mount RCD all of the conductors must pass through the CT.
5. Loosen the load side lugs and remove the wires from the terminals as much as possible preserve the wire bending radius and routing.
6. Pull the conductors free from the circuit breaker and through the CT
7. Remove the (4) mounting screws from the CT and replace the unit.
8. Identify the RCD module (1) and verify that the part number in the FRU kit is an exact match of the part number removed from the panel.
9. Prior to loosening the terminations on the RCD module observe the configuration, wire routings, RCD settings, wire labels, and terminations for each terminal.
10. The RCD module is DIN rail mounted and is secured by a tension clamp. Insert a screw driver into the mounting clamp tab and gently lift up to release the RCD module from the DIN rail.
11. Replace the RCD module provided in the FRU kit. When seated on the DIN rail push the mounting clamp tab down to secure the RCD module.



12. Replace each termination according to the notes taken in step 9. Refer to the component installation document for torque and tightening specifications.
13. Using the existing settings of the RCD module as a template, set the parameters of the RCD module provided in the FRU kit.
14. Route the conductors back through the CT and tighten the terminations according to the torque values on the schematic fixed to the cover of the panel door.
15. Prior to energizing the MDP, consult the schematic fixed to the cover of the panel door and verify all connections are terminated properly, ALL CONDCTORS ARE ROUTED THROUGH THE CT, and that the RCD module settings are set properly.

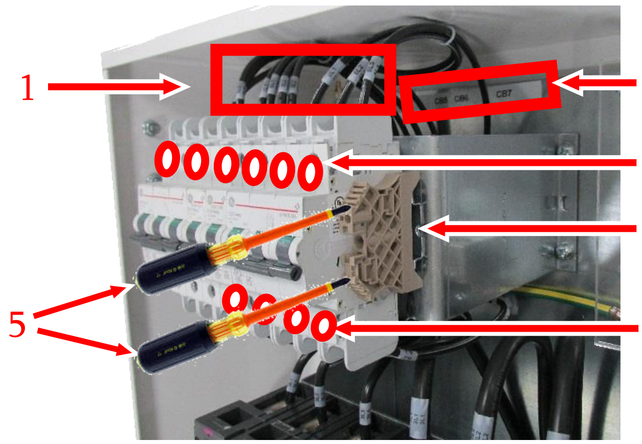
MINIATURE AND CONTROL CIRCUIT BREAKER FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the circuit breaker screws.

2. Prior to performing any work observe the circuit breaker configuration (6), wire labels and location (1).



3. Loosen the screws from the terminals on the top and bottom of the circuit breaker (3 &4) and remove the wires.
4. Miniature Circuit Breakers are most often DIN rail mounted and secured by an end clamp (2). The end clamp has 1 tension screw on the top. Loosen, the tension screws (5), then slide the end clamp to the nearest edge until it is removed from the DIN rail.
5. With the end clamp removed slide the circuit breaker(s) to the nearest edge of the DIN rail. As each circuit breaker is removed, discretely identify them so that they can be reinstalled in the correct sequence so that the device tags (6) properly align with the corresponding circuit breaker.
6. Verify the circuit breaker part number provided in the FRU kit is an exact match of the components removed from the panel.
7. Install the replacement circuit breaker(s) in the reverse order in which they were removed and verify that they properly align with the device tags (6) above each component.
8. Slide the end clamp (2) onto the DIN rail. Each of the circuit breakers should be installed so that there are no gaps between components. Fasten the end clamp screws (5) and be certain not to over tighten.
9. Route the wires to the circuit breaker preserving the original path to the terminals.
10. Refer to the instruction and specifications document provided with each component and fasten each termination according to the torque settings specified for each circuit breaker.

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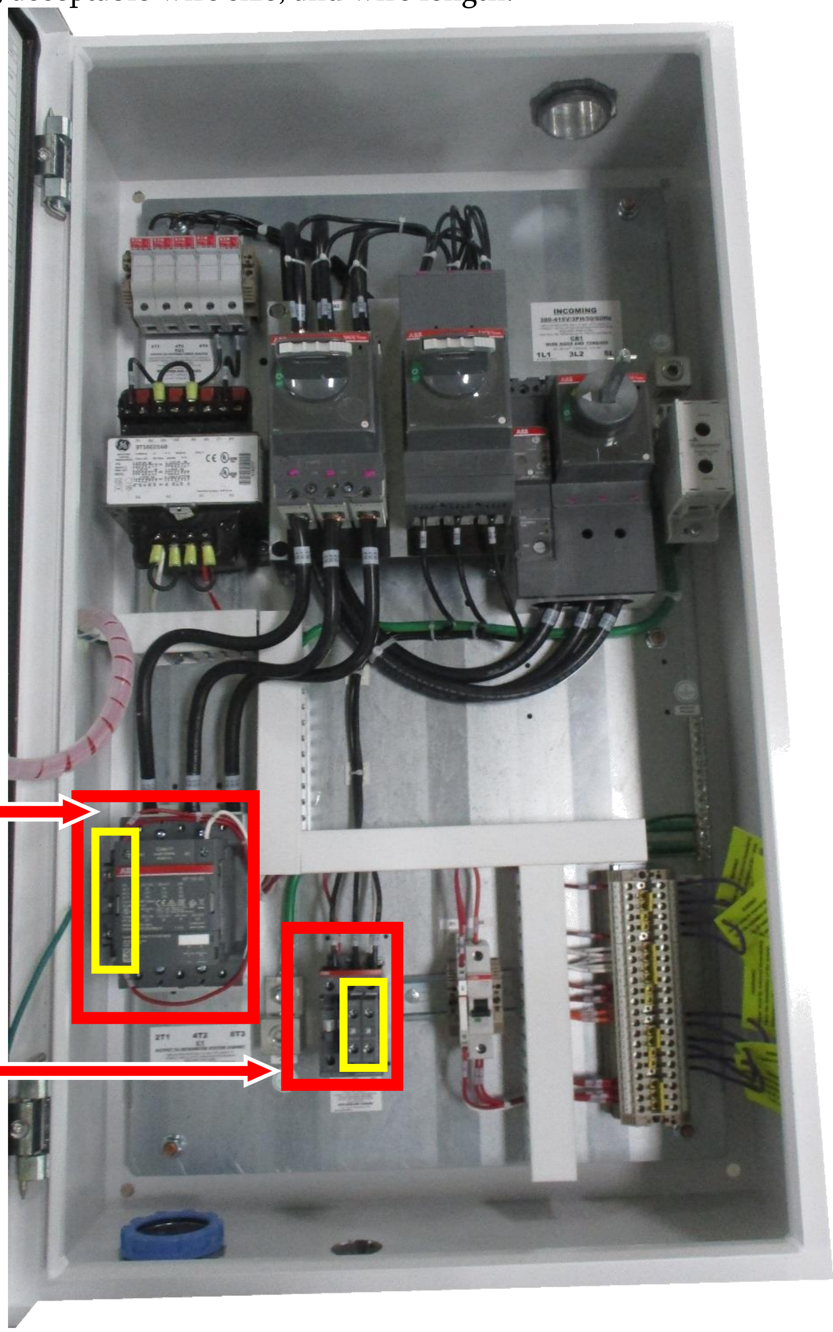
CONTACTORS FRU's

Contactors are mounted within the panel in 2 distinct applications. Larger contactors are mounted to the sub panel whereas mini contactors are most often DIN rail mounted. These components are commonly designed to re-energize automatically once power is restored to the main circuit breaker or the control circuit via an auxiliary contact. Auxiliary contacts come in both NORMALLY OPEN and NORMALLY CLOSED default positions and can be side or front mounted. Installation specifications and requirements vary by manufacturer.
A separate section detailing the replacement of the auxiliary contact is provided in this document. Like any electrical component a contactor should be treated as though it were energized at all times. Always adhere to the requirements specified for torque values, acceptable wire size, and wire length.

Sub Panel Mounted
Contactor with auxiliary
contact

DIN Rail Mounted
Contactor with auxiliary
contacts



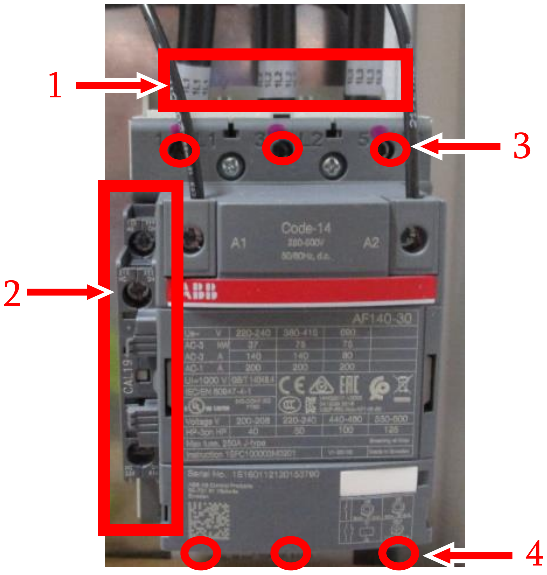
CONTACTOR - SUBPANEL MOUNTED FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

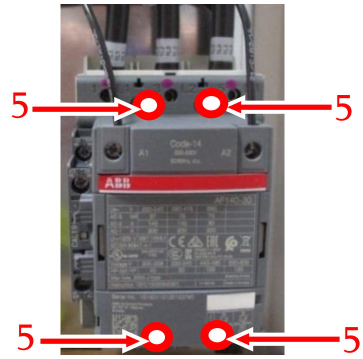
1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the contactor terminals.

2. Prior to performing any work observe the contactor configuration; wire labels and location (1) and auxiliary contact accessories (2).



3. Loosen, the terminal screws on the top and bottom of the contactor (3 & 4) and remove the wires from the component. Preserve the wire bend radius and routing as much as possible.

4. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the mounting bolts (5).
5. Remove the four mounting bolts (5) from the contactor. DO NOT DISCARD.



6. Remove the contactor from the sub panel and verify the part number(s) provided in the FRU kit is an exact match of the component(s) removed from the panel.

For contactors equipped with an auxiliary contact please see the following page

7. Mount the contactor provided in the FRU kit and then install the mounting bolts from step 4.
8. As much as possible preserve the wire bending radius as noted in step 3 and then land the appropriate wires in the designated terminals on the contactor.
9. Refer to the instruction and specifications document provided with each component and fasten each termination according to the specified torque settings.

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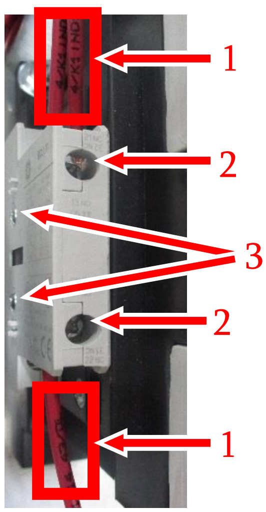
CONTACTOR – AUXILIARY CONTACT FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the contact terminals.

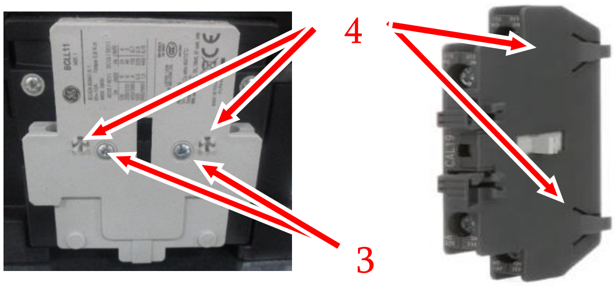
2. Prior to performing any work observe the auxiliary contact configuration; wire labels and location (1), terminal screws (2), and the mounting guides and/or screws (3).



3. Loosen the terminal screws on the top and bottom of the auxiliary contact (2) and then remove the wires from the terminals.
4. Refer to the instruction manual provided with the auxiliary contact to identify the proper tool(s) required to loosen any mounting screws (3).

5. DO NOT DISCARD. Front mounted contacts (5) are secured by mounting guides and can easily be removed by sliding the contact up and away from the contactor frame.
6. Once the contact is removed from the contactor verify the part number(s) provided in the FRU kit is an exact match of the component removed from the panel.

7. Mount the contact provided in the FRU kit. Most contacts provide mounting guides (4) that extend from the frame and facilitate proper seating. Fasten the auxiliary contact to the contactor using the mounting screws if included (3).



8. As much as possible preserve the wire routing as noted in step 2 and then land the appropriate wires in the designated terminals on the contact.
9. Refer to the instruction and specifications document provided with the auxiliary contact and fasten each termination according to the specified torque settings.

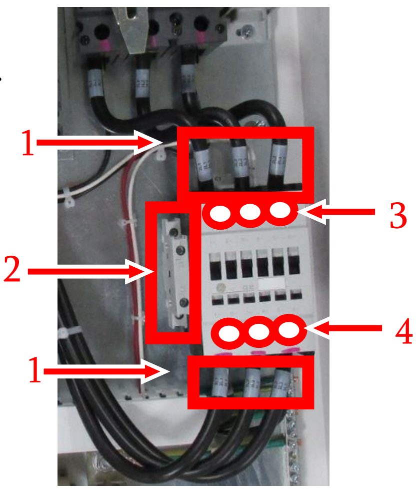
CONTACTOR – DIN RAIL MOUNTED FRU's

DANGER

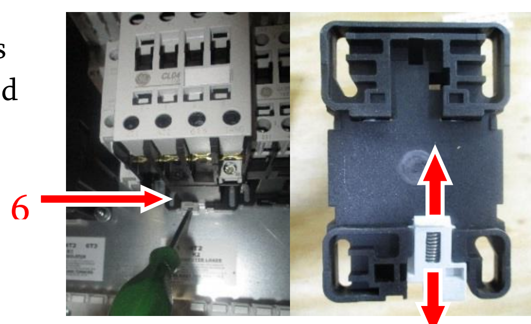
Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the contactor terminals.

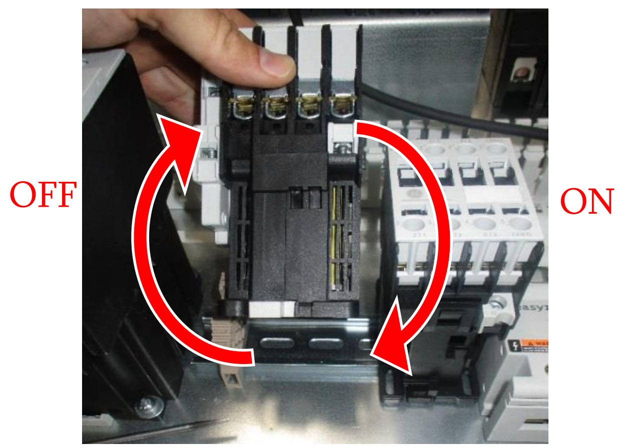
2. Prior to performing any work observe the contactor configuration; wire labels and location (1), and auxiliary contact accessory (2).
3. Loosen, the terminal screws on the top and bottom of the contactor (3 & 4), and remove the wires from the component, and preserve the wire bend radius and routing as much as possible.



4. DIN rail contactors are secured with a spring-loaded tension clamp (6) located on the back of the contactor.



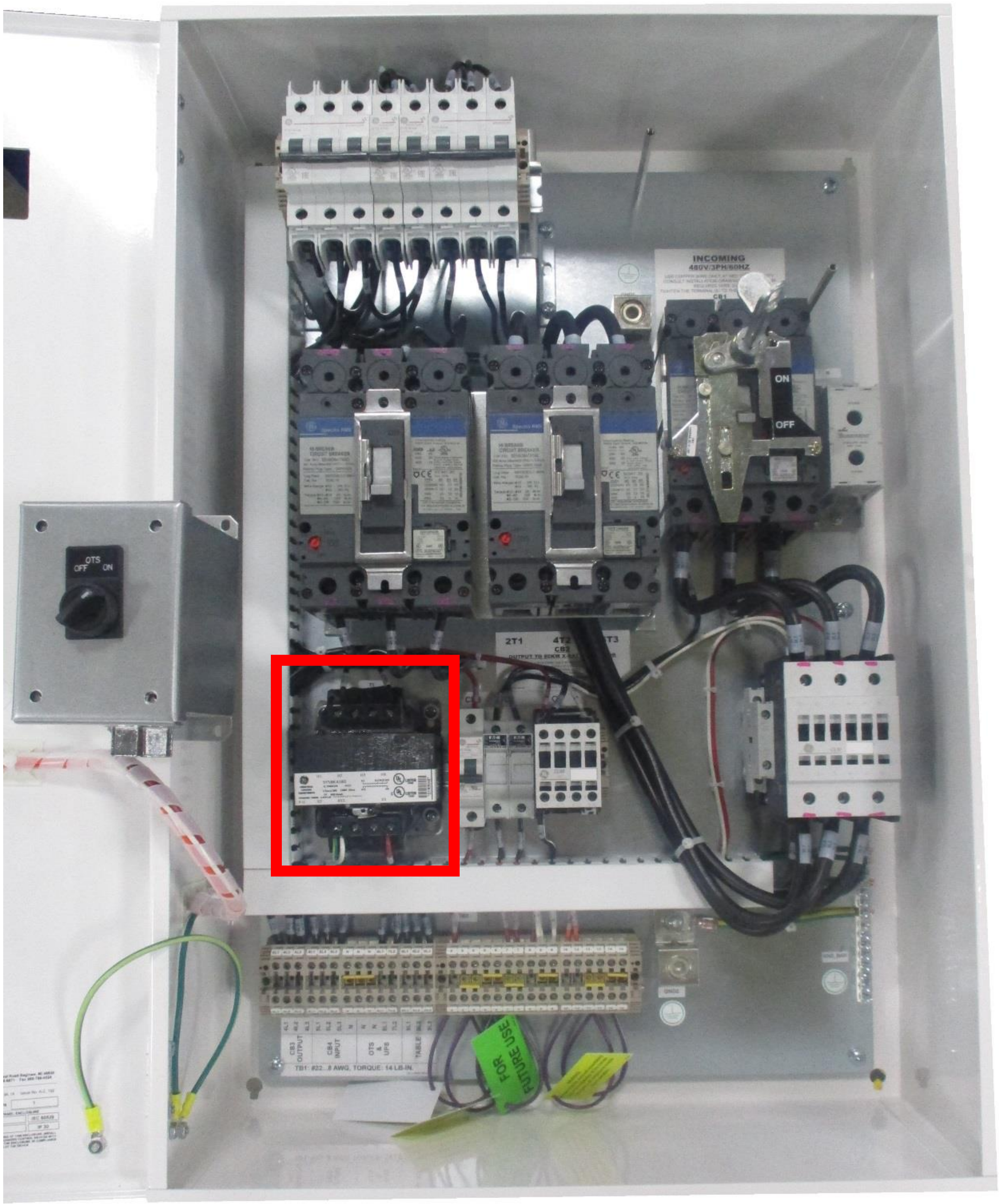
5. Release the tension clamp by inserting a standard screw driver head into the tab (6) as shown in step 4 and pulling the tab downwards, away from the frame of the contactor.
6. To remove the contactor from the DIN rail, lift up on the contactor from the bottom of the frame so that the spring-loaded tension clamp moves outward, up, and away from the subpanel as shown below.
7. Verify the part number(s) provided in the FRU kit is an exact match of the component(s) removed from the panel.



8. Mount the contactor provided in the FRU kit on the DIN rail by first opening the spring-loaded tension clamp, then aligning the top of the contactor with the DIN rail first. Secure the bottom of the contactor frame to the DIN rail last and push the tab for the tension clamp in to lock the contactor in place.
10. As much as possible preserve the wire bending radius as noted in step 3 and then land the appropriate wires in the designated terminals on the contactor.
11. Refer to the instruction and specifications document provided with each component and fasten each termination according to the specified torque settings.

TRANSFORMER FRU's

Transformers within the main disconnect panel are typically fed from the load side of the main circuit breaker or contactor. Always adhere to the requirements specified for torque values, acceptable wire size, and wire length.



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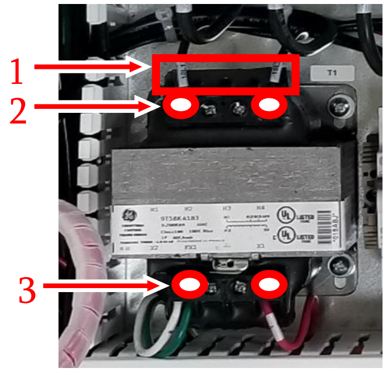
TRANSFORMER FRU's

DANGER

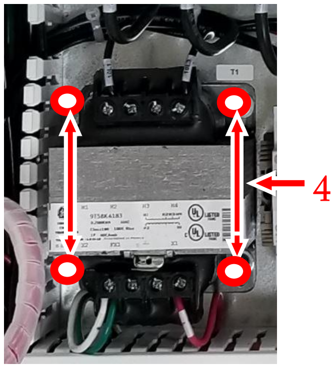
Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the transformer terminals.

2. Prior to performing any work observe the transformer configuration; wire labels and wire locations (1) as well as the primary (2) and secondary terminations (3)



3. Loosen, the terminal screws on the primary (2) then secondary (3) terminals of the transformer and remove the wires from the component. Preserve the wire bend radius and routing as much as possible.
4. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the mounting bolts (4).



5. Verify the transformer part number provided in the FRU kit is an exact match of the components removed from the panel.
6. Mount the contactor provided in the FRU and then install the mounting bolts from step 4.
7. As much as possible preserve the wire bending radius as noted in step 3 and then land the appropriate wires in the designated terminals on the transformer.
8. Refer to the instruction and specifications document provided with the transformer and fasten each termination according to the torque settings accordingly.

POWER SUPPLY FRU's

Power supplies within the main disconnect panel are typically DIN rail mounted and tapped from the load side of the main circuit breaker. Single pole fuses are commonly installed ahead of the incoming power. Always adhere to the requirements specified for torque values, acceptable wire size, and wire length.

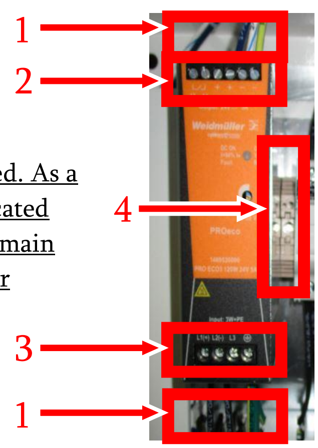
POWER SUPPLY FRU's

DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the power supply terminals.
2. Prior to performing any work observe the power supply configuration; wire labels and wire locations (1).

<u>***Warning: The incoming power may be tapped ahead of the main circuit breaker. Power may be present even if the main circuit breaker is de-energized. As a precaution open the fuses located between the load side of the main circuit breaker and the power supply.</u>



3. Loosen the terminal screws from on the top and bottom of the power supply (2 & 3) and remove the wires from the terminals.
4. Power supplies are secured to DIN rail with a spring-loaded tension clamp (5) located on the back of the power supply frame.
5. Release the tension clamp by inserting a standard screw driver head into the tab (6) and pulling the tab upwards, away from the frame of the power supply.

6. To remove the supply from the DIN rail, lift up on from the bottom of the power supply frame so that the spring-loaded tension clamp moves up, and away from the subpanel.
7. Verify the power supply part number provided in the FRU kit is an exact match of the components removed from the panel.
8. Mount the power supply so that the bottom cleats are fixed to the DIN rail first, then align the top of the power supply with the DIN rail and push until the spring-loaded tension clamp snaps onto the DIN rail and the power supply is secure.
9. As much as possible preserve the wire routing as noted in step 2 and then land the appropriate wires in the designated terminals on the power supply.
10. Refer to the instruction and specifications document provided with the power supply and fasten each termination according to the specified torque settings.

PILOT/INDICATOR LIGHTS, SELECTOR SWITCH, AND EPO FRU's

Pilot devices provide a visual indication of the presence of power, component status, and provide a means to disconnect and energize the components within the panel. In the event an indicator light is not illuminated or the selector switch is set to the OFF position the panel should always be treated as though it were energized at all times. The various types of applications shown in the example below are secured by a locking collar that allows the component to be fastened to the knock out in the enclosure's door.

PILOT/INDICATOR LIGHTS, SELECTOR SWITCH, AND EPO FRU's

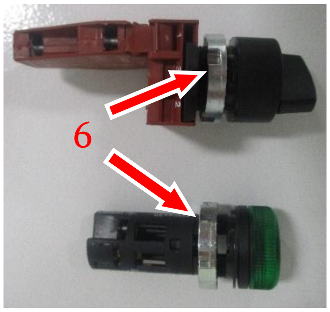
DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. For replacement of each component refer to the instruction manual provided and identify the proper tool(s) required to loosen the contact block terminals.

1

2. Prior to performing any work observe the configuration and connections of the device, wire labels and wire locations.
3. Loosen the terminal screws (1) and remove the wires from the terminals.
4. Remove the locking collar (6) from the threaded portion of the component.



6

5. Verify the part numbers provided in the FRU kit is an exact match of the components removed from the panel.
6. Insert the indicator light or selector switch through the knock-out on the enclosure door and thread the locking collar onto the indicator light or selector switch as shown in step 4.
7. As much as possible preserve the wire routing as noted in step 2 and then land the appropriate wire in the designated terminals (1).
8. Refer to the schematic fixed to the panel cover and verify that all terminations are torqued and landed according to the schematic.

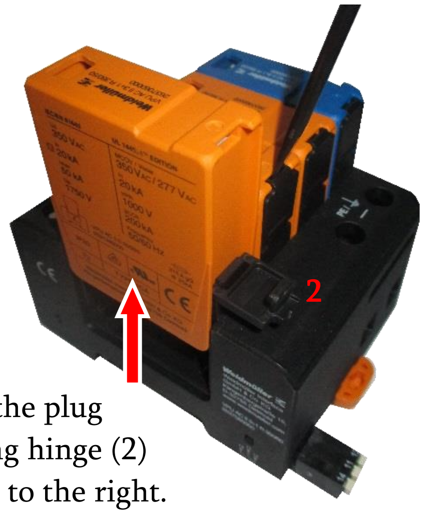
SURGE PROTECTIVE DEVICE (SPD) FRU's

Surge Protective Devices or SPD's are intended to divert or limit the damage caused by transient voltage. SPD's are 'single use' devices that provide a visual status indicator that changes from GREEN to RED when the device needs to be replaced.

SURGE PROTECTIVE DEVICE (SPD) FRU's

DANGER

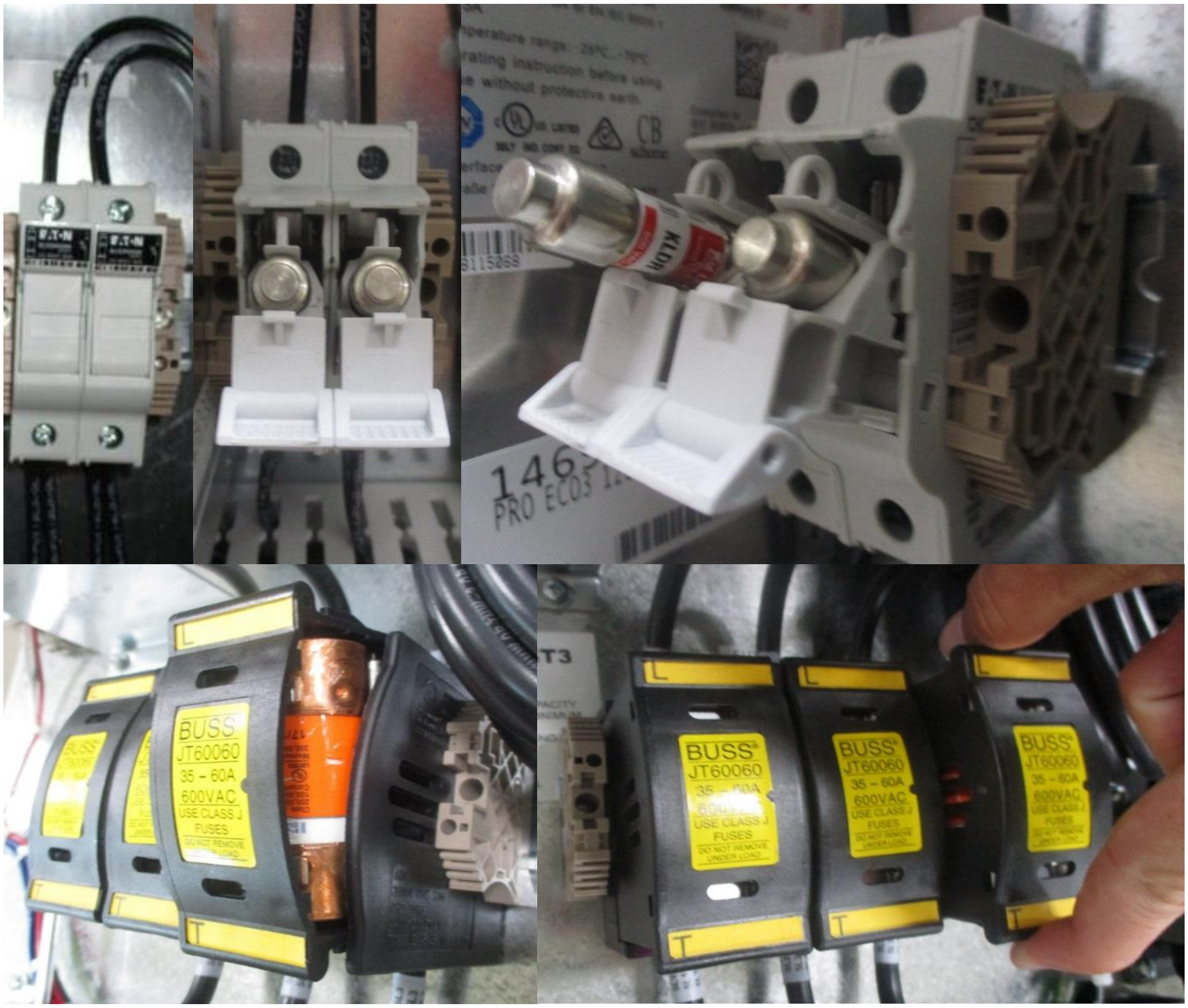
Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.



1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to replace the device.
2. Prior to performing any work observe the configuration of the device, wire routing, and terminations.
3. The SPD comes with a status indicator window (1) for visual inspection of the device. A GREEN window represents the normal functional operation of the device.
4. Each of the varistors can be manually removed and are easily field replaceable by pulling the varistor directly out and away from the plug
5. Depending on manufacturer a locking hinge (2) may need to be disengaged as shown to the right.

TIME DELAY CURRENT LIMITING FUSE FRU's

Time Delay Fuses and fuse holders are current limiting devices that provide protection to the control circuit components. These fuses range in ampere based on the requirements of each design. The fuses on both sides of the circuit must be in working order for the panel to operate properly. Not all fuses are interchangeable. The peak let-through current and failure time characteristics of each fuse is different for each manufacturer even though the ampere ratings may be the same. When replacing any fuse, it should only be done so after consulting the replacement fuse chart located on the schematic.



TIME DELAY CURRENT LIMITING FUSE FRU's

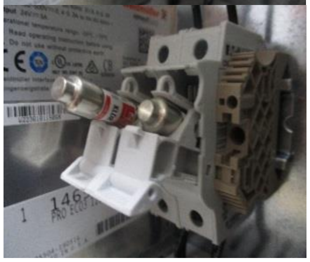
DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. For replacement of each fuse refer to the schematic located on the inside of the enclosure door and consult the approved manufacturers replacement fuse chart. Do not install fuses that are not approved.

2. Pull outward on the fuse holder tab until the fuses are exposed.

3. Remove the fuse(s) and replace them only with approved replacement manufacturers part paying attention to the orientation of the fuse as it is inserted into the fuse holder.

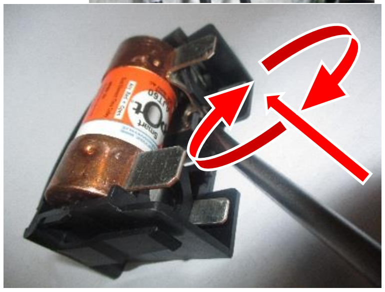


4. Push the fuse holder back into place and verify that the proper voltage is present.

5. For larger fuses, such as J class fuses follow steps 6-11.

6. Remove the fuse holder by manually pulling the fuse away from the panel.

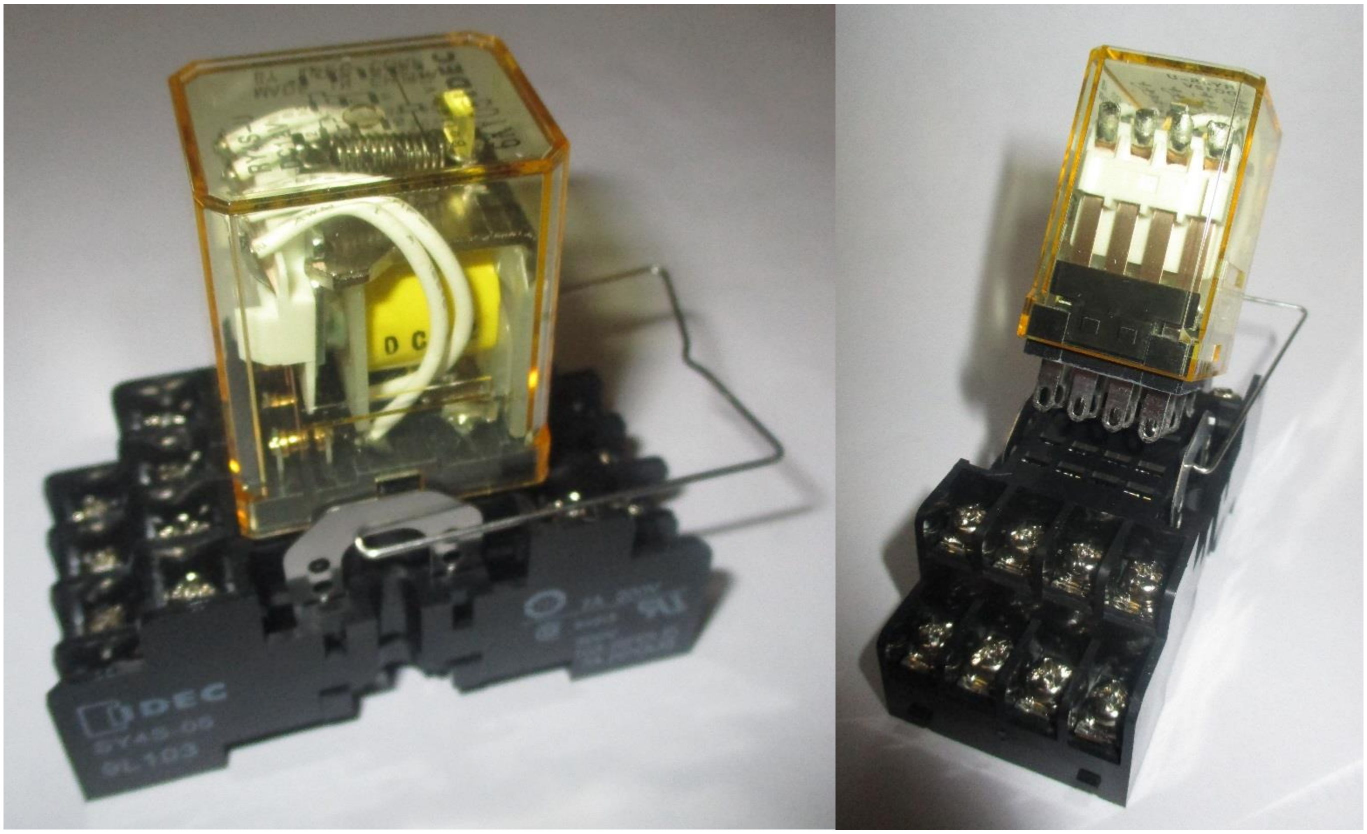
7. Using a large standard screw driver insert the head of the screw driver into the outer most corner of the fuse holder frame and rotate the screw driver clockwise. Perform the same steps in the opposite end of the fuse holder. This method will disengage the fuse from the holder.



8. Replace the fuse only with part numbers located on the fuse replacement chart located on the schematic.
9. When inserting the replacement fuse, do so, so that both ends of the fuse are pushed in evenly.
10. Insert the fuse into the fuse holder and test the panel.

CONTROL RELAY FRU's

Control relays are self-contained coils and relays that close a contact when energized. The control relay comes assembled as one piece with the terminal block and relay coil. Removing the relay coil from the terminal block is the easiest method to replacement as the terminals do not have any mechanical elements and rarely fail.

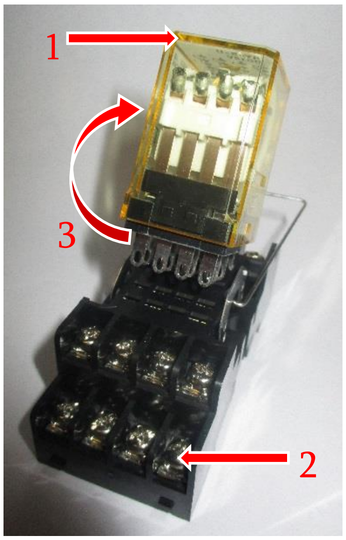


CONTROL RELAY FRU's

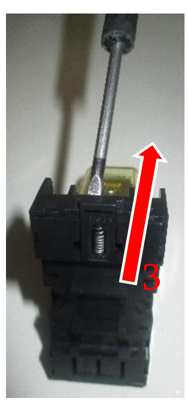
DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

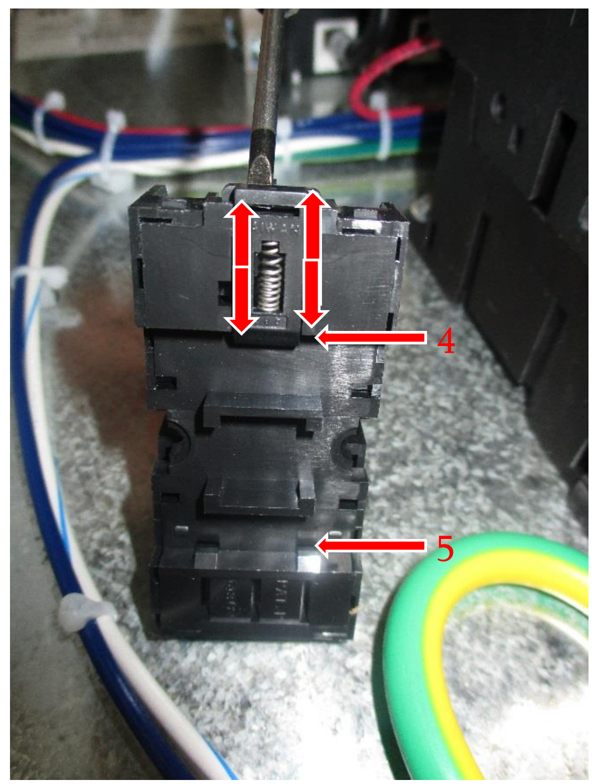
1. For replacement of each component refer to the instruction manual provided and identify the proper tool(s) required to loosen the control relay.
2. Prior to performing any work observe the control relay configuration; wire labels and wire locations.



3. Remove the remove the coil and relay capsule (1) from the terminal block (2) by manually pulling outward and away from the panel.
4. Align the relay pins (3) with the terminal block orientation and plug the relay coil into the designated slots.
5. In the event that the terminal block needs to be replaced observe the control relay configuration; wire labels and wire locations prior to performing any work.
6. Loosen the terminal screws and remove the wires from the terminals.
7. The terminal block comes DIN rail mounted and can be removed by inserting a standard screw driver into the tab (3) for spring-loaded retention clamp and pulling away from the component until the terminal block is disengaged from the DIN rail.



8. The terminal block can be installed onto the DIN rail by aligning its bottom cleat (4) with the bottom edge of the DIN rail then pushing the upper cleat (5) of the terminal block into the upper edge of the DIN rail until the component snaps into place.



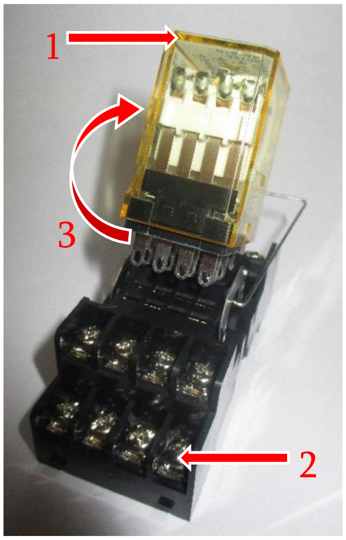
9. Land the wires on the terminals noted in step 2.

CONTROL RELAY FRU's

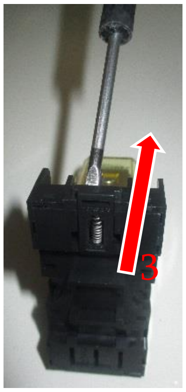
DANGER

Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

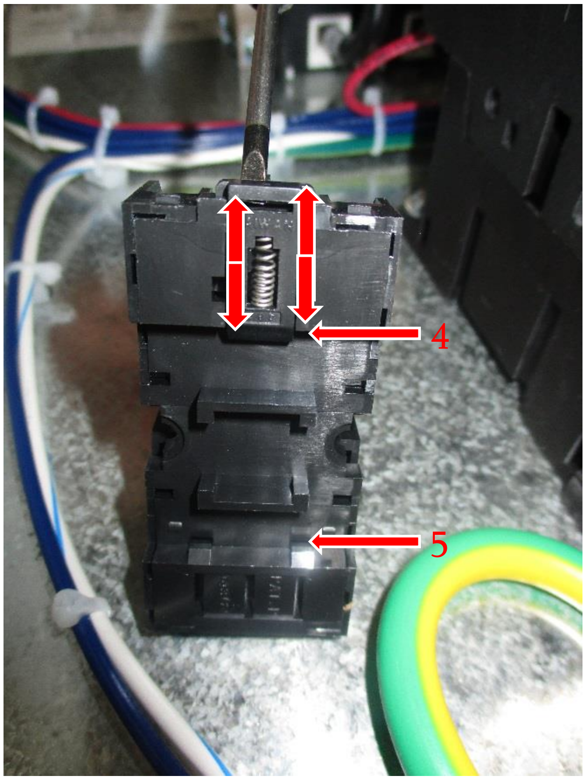
10. For replacement of each component refer to the instruction manual provided and identify the proper tool(s) required to loosen the control relay.



11. Prior to performing any work observe the control relay configuration; wire labels and wire locations.
12. Remove the remove the coil and relay capsule (1) from the terminal block (2) by manually pulling outward and away from the panel.
13. Align the relay pins (3) with the terminal block orientation and plug the relay coil into the designated slots.
14. In the event that the terminal block needs to be replaced observe the control relay configuration; wire labels and wire locations prior to performing any work.



15. Loosen the terminal screws and remove the wires from the terminals.
16. The terminal block comes DIN rail mounted and can be removed by inserting a standard screw driver into the tab (3) for spring-loaded retention clamp and pulling away from the component until the terminal block is disengaged from the DIN rail.
17. The terminal block can be installed onto the DIN rail by aligning its bottom cleat (4) with the bottom edge of the DIN rail then pushing the upper cleat (5) of the terminal block into the upper edge of the DIN rail until the component snaps into place.



18. Land the wires on the terminals noted in step 2.

INTERFACE RELAY FRU's

Interface relays are isolated coils that improve the reliability in control signaling. DIN rail mounted the control relay comes assembled as one piece with the terminals and relay coil.

INTERFACE RELAY FRU's

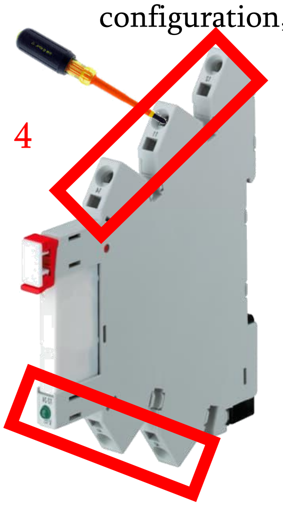
DANGER

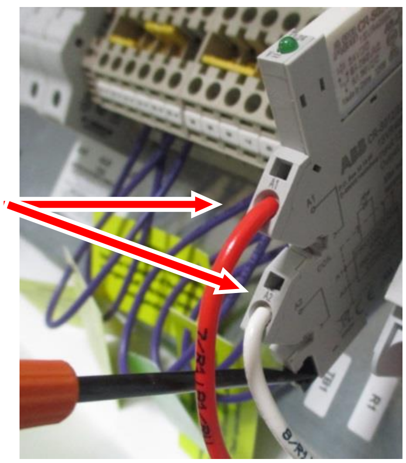
Only authorized and qualified personnel are allowed to service the MDP. Prior to performing any service or maintenance, safely and properly de-energize the main disconnect panel, then Lock Out and Tag Out the panel.

1. Refer to the instruction manual provided with each component and identify the proper tool(s) required to loosen the circuit breaker screws.

2. Prior to performing any work observe the relay configuration, wire labels and location (1).

4





3. Loosen the screws from the terminals on the top and bottom of the relay (3 &4) and remove the wires.
4. DIN rail mounted and secured by an end clamp (2). The end clamp has 1 tension screw on the top. Loosen, the tension screws (5), then slide the end clamp to the nearest edge until it is removed from the DIN rail.
5. With the end clamp removed slide the circuit breaker(s) to the nearest edge of the DIN rail. As each circuit breaker is removed, discretely identify them so that they can be reinstalled in the correct sequence so that the device tags (6) properly align with the corresponding circuit breaker.
6. Verify the circuit breaker part number provided in the FRU kit is an exact match of the components removed from the panel.
7. Install the replacement circuit breaker(s) in the reverse order in which they were removed and verify that they properly align with the device tags (6) above each component.
8. Slide the end clamp (2) onto the DIN rail. Each of the circuit breakers should be installed so that there are no gaps between components. Fasten the end clamp screws (5) and be certain not to over tighten.
9. Route the wires to the circuit breaker preserving the original path to the terminals.
10. Refer to the instruction and specifications document provided with each component and fasten each termination according to the torque settings specified for each circuit breaker.

LOCK-OUT / TAG-OUT

"Lockout/tagout" refers to the specific practices and procedures to safeguard individuals from the unexpected energization or startup of equipment and the release of hazardous energy during service or maintenance activities. This requires, a qualified and designated individual turns off, then disconnects the equipment from its energy source(s) before performing service or maintenance, and then lock-out and tag-out the service to the equipment to prevent the release of hazardous energy. Steps to verify that the energy has been isolated effectively should always be completed prior to performing and maintenance or service.

1. Prepare for shutdown - locate, identify, and confirm the location of the facility circuit(s) feeding the main disconnect panel.
2. Shut down the main disconnect panel by manually turning off the circuit breaker(s) exposed through the main disconnect panels enclosure door.
3. Disconnect or isolate the main disconnect panel from the facility energy source(s).
4. Apply the Lock-out Tag-out device(s) to the energy-isolating device/facility circuit breaker feeding the panel.
5. Every main disconnect panel is designed with an OSHA approved main circuit breaker rotary disconnect handle or lever disconnect handle Lock-out Tag-out device. Branch circuit breakers are also commonly equipped with OSHA approved Lock-out Tag-out hasps. The main circuit breaker should be Locked-out and Tagged-out whenever possible as it offers the most protection against injury or death. When the main circuit breaker is unable to be de-energized the branch circuit breaker protective Lock-out Tag-out hasp needs to be utilized at the minimum. Service or maintenance should never be performed on equipment, panels, or components that are energized.